Medical Assistant Project

Natural Language Processing with Generative AI: Medical Assistant

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Executive Summary

Insights

- Delays or inaccuracies in accessing medical information can directly impact patient outcomes, highlighting the need for tools that enhance precision and speed in diagnostics and treatment planning.

- There is information overload in this industry and healthcare professionals are faced with the challenge in managing the forever growing volume of medical data, leading to difficulties in accessing relevant and up to date information.

- There is an opportunity to improve the consistency and quality of care by providing a centralized, easily accessible source of trusted medical information from renowned manuals and research.

Executive Summary

Recommendations

- A RAG-based AI solution is necessary to retrieve context aware, accurate and timely information from the medical manuals.
- Establish a robust pipeline for regularly updating the RAG system with the latest versions of medical manuals, research papers, and clinical guidelines.
- Develop a user friendly interface so all healthcare professionals can quickly formulate queries and get a quality response in under a minute.

Business Problem Overview and Solution Approach

- There is mass medical data available to healthcare professionals. Every day they are challenged with a high pace environment and many patients. They often encounter information overload and need assistance finding the information they need in order to provide the best care possible.

- The solution to this issue is to create a RAG based AI solution that can retrieve the information they need just by asking it questions. We will create a model that is feasible and effective when it comes to providing the best answers that will result in the best care for the patients.

Question Answering using LLM

Model Name and Parameters & Question 1

- The name of the model we will use for this project is TheBloke/Mistral-7B-Instruct-v0.2-GGUF
- Our max Tokens = 2000, temperature = 0, top_p=0.95, and top_k=50

The first question we have is: What treatment options are available for managing hypertension?

This was the answer we received:

Hypertension, or high blood pressure, is a common condition that can increase the risk of various health problems such as heart disease, stroke, and kidney damage. The good news is that there are several effective treatment options available to help manage hypertension and reduce the risk of complications. Here are some of the most commonly used treatments:Lifestyle modifications: Making lifestyle changes is often the first line of defense against hypertension. This may include eating a healthy diet rich in fruits, vegetables, whole grains, and lean proteins; limiting sodium intake; getting regular physical activity; maintaining a healthy weight; and reducing stress through techniques such as meditation or deep breathing exercises.Medications: If lifestyle modifications alone are not enough to control hypertension, medications may be necessary. There are several classes of drugs used to treat hypertension, including diuretics, beta blockers, ACE inhibitors, calcium channel blockers, and ARBs (angiotensin receptor blockers). These medications work by reducing the amount of blood your heart pumps and/or the amount of fluid in your body, which can help lower blood pressure. Combination therapy: In some cases, a combination of lifestyle modifications and medications may be necessary to effectively manage hypertension. Your healthcare provider may recommend a specific combination of treatments based on your individual needs and health conditions. Alternative therapies: There are also several alternative therapies that have been shown to help lower blood pressure, although more research is needed to fully understand their effectiveness. These include acupuncture, meditation, yoga, and certain herbs and supplements such as garlic, magnesium, and coenzyme Q10.Lifestyle modifications for specific conditions: In some cases, hypertension may be caused by an underlying condition such as kidney disease or sleep apnea. In these cases, treating the underlying condition may help improve blood pressure. For example, managing kidney disease with dialysis or a transplant, or using a continuous positive airway pressure (CPAP) machine to treat sleep apnea can help lower blood pressure. t's important to note that everyone's body responds differently to treatment, so what works best for one person may not work as well for another. Your healthcare provider will work with you to determine the most effective treatment plan based on your individual needs and health conditions.

Question Answering using LLM

Question 1 Analysis

The model described what Hypertension is and then what it can lead to. It then lists treatment options that the healthcare professional can use to manage hypertension and reduce the risk of complications. It then goes in details on lifestyle changes that can reduce hypertension and alternative therapies such as acupuncture, meditation, yoga,. This is an effective answer that will help the patient receive the best health advice.

Question Answering using LLM

Query 1 & Analysis

Query 1: What is the protocol for managing Sepsis in a Critical care unit?

Answer:

Sepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit: Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis. ABCs: Ensure airway patency, adequate breathing, and circulatory support. Provide high-flow oxygen via a non-rebreather mask or endotracheal tube if necessary. Initiate intravenous fluids to maintain adequate blood pressure and organ perfusion. Antibiotics: Administer broad-spectrum antibiotics as soon as possible based on the suspected source of infection and local microbiology data. Consider obtaining cultures before administering antibiotics if clinically feasible. Source control: Identify and address the source of infection, such as drainage of an abscess or debridement of necrotic tissue. Hemodynamic support: Provide adequate fluid resuscitation to maintain adequate organ perfusion. Consider using vasopressors if necessary to maintain mean arterial pressure (MAP) greater than 65 mmHg. Inotropic support: Use inotropes to maintain cardiac output and tissue perfusion if necessary. Renal replacement therapy: Consider initiating renal replacement therapy if there is evidence of severe sepsis or septic shock, such as persistent metabolic acidosis, anuria, or oliguria. Glucose control: Maintain strict glucose control to prevent hyperglycemia, which can worsen inflammation and tissue damage. Aim for a blood glucose level between 80 and 150 mg/dL. Corticosteroids: Consider using corticosteroids in patients with septic shock who do not respond to fluid resuscitation and vasopressors. Other supportive measures: Provide other supportive measures as needed, such as mechanical ventilation, vasoactive medications, and nutritional support. Monitoring: Close monitoring of vital signs, laboratory values, and clinical status is essential to ensure effective treatment and timely intervention if necessary. Early goal-directed therapy (EGDT): EGDT is a protocolized approach to managing sepsis that involves early recognition, aggressive fluid resuscitation, and goal-directed therapy to optimize tissue perfusion and oxygenation. This approach has been shown to improve outcomes in patients with severe sepsis or septic shock. Multidisciplinary team: A multidisciplinary team, including critical care physicians, nurses, pharmacists, and other healthcare professionals, is essential for effective management of sepsis in a critical care unit. Regular communication and collaboration among team members are crucial to ensure timely and appropriate interventions.

Analysis

The response describes what Sepsis is and goes through many different options to help manage it. This gives the medical professional the best advice to provide care for their patient.

Question Answering using LLM

Query 2 & Analysis

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should follow to treat it?

Answer: Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs include: 1. Abdominal pain: The pain typically starts as a mild discomfort around the navel or in the lower right abdomen. It may then move to the right lower quadrant and become more severe over time. The pain is often described as sharp, crampy, or constant. 2. Loss of appetite: People with appendicitis may lose their appetite due to abdominal discomfort or nausea. 3. Nausea and vomiting: Vomiting is a common symptom of appendicitis, especially in the later stages of the condition. 4. Fever: A fever of 100.4°F (38°C) or higher may be present in some cases of appendicitis. 5. Diarrhea or constipation: Some people with appendicitis experience diarrhea, while others have constipation. 6. Rebound tenderness: When pressure is applied to the abdomen and then released, there may be pain or tenderness in the right lower quadrant, which is a sign of peritonitis (inflammation of the lining of the abdominal cavity). 7. Guarding: People with appendicitis may involuntarily contract their abdominal muscles to protect the area from pressure or pain. Appendicitis cannot be cured via medicine alone, as the inflammation and swelling of the appendix can obstruct the blood flow, leading to tissue death and potential rupture. If left untreated, a ruptured appendix can lead to peritonitis, which is a serious and potentially life-threatening condition. Therefore, surgery is the standard treatment for appendicitis. The surgical procedure used to treat appendicitis is called an appendectomy. During this procedure, the surgeon makes an incision in the abdomen and removes the inflamed appendix. The two most common types of appendectomies are open appendectomy and laparoscopic appendectomy: 1. Open appendectomy: In this procedure, the surgeon makes a single large incision (usually about 2-3 inches long) in the abdomen to access the appendix. The appendix is then removed, and the incision is closed with sutures or staples. 2. Laparoscopic appendectomy: In this minimally invasive procedure, the surgeon makes several small incisions (usually less than 1 inch long) in the abdomen to insert a laparoscope and other surgical instruments. The appendix is then removed using these instruments, and the incisions are closed with sutures or staples. Both open and laparoscopic appendectomies have their advantages and disadvantages, and the choice of procedure depends on various factors, including the patient's age, overall health, and the severity of the appendicitis. In general, laparoscopic appendectomy is preferred when possible due to its smaller incisions, faster recovery time, and lower risk of complications. However, open appendectomy may be necessary in certain cases, such as when the appendix has ruptured or when there are other complicating factors. After surgery, patients typically need to stay in the hospital for a few days to recover. They may receive pain medication, intravenous fluids, and antibiotics to prevent infection. Once discharged from the hospital, patients can gradually return to their normal activities as they heal. It is important to follow the doctor's instructions carefully during recovery to ensure a successful outcome.

Analysis: The response does go in detail about what Appendicitis is and describes common signs. It tells you what kind of surgery is needed and goes into a lot of detail

Question Answering using LLM

Query 3 & Analysis

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Answer: Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes. The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certain medications. There are several treatments that have been shown to be effective in addressing sudden patchy hair loss: 1. Corticosteroids: These are anti-inflammatory drugs that can help reduce inflammation and suppress the immune system's attack on the hair follicles. They can be applied topically or taken orally, depending on the severity of the condition. 2. Minoxidil: This is a medication that has been shown to promote hair growth in some people with alopecia areata. It works by increasing blood flow to the affected area and stimulating hair follicles to produce new hairs. 3. DHT blockers: Dihydrotestosterone (DHT) is a hormone that can contribute to hair loss. Some people with alopecia areata have high levels of DHT in their scalp. Finasteride and dutasteride are medications that can block the production of DHT, which may help promote hair growth. 4. Immune system modulators: These drugs work by suppressing the immune system's attack on the hair follicles. Examples include methotrexate, mycophenolate mofetil, and cyclosporine. 5. Hair transplantation: In some cases, hair transplantation may be an option for people with alopecia areata who have large bald patches that don't respond to other treatments. This involves taking healthy hairs from another part of the scalp or body and transplanting them into the affected area. It's important to note that there is no cure for alopecia areata, and not all treatments work for everyone. The effectiveness of treatment can also depend on the severity and location of the bald patches. If you're experiencing sudden patchy hair loss, it's best to speak with a healthcare professional or dermatologist to determine the underlying cause and discuss your treatment options.

Analysis: The response answers the full question and says there is not a known exact cause. It then goes into the different treatments and solution for the patients who have alopecia aerata.

Question Answering using LLM

Query 4 & Analysis

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Answer: A person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are some common treatments that may be recommended: 1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical attention as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening conditions. 2. Medications: Depending on the symptoms, healthcare professionals may prescribe medications to manage various conditions associated with a brain injury, such as pain, swelling, seizures, or infections. 3. Surgery: In some cases, surgery may be necessary to remove blood clots, repair skull fractures, or relieve pressure on the brain. 4. Rehabilitation: Rehabilitation is an essential component of treatment for brain injuries. It may include physical therapy, occupational therapy, speech and language therapy, and cognitive rehabilitation to help individuals regain lost skills and improve overall function. 5. Assistive devices: Depending on the extent of the injury, assistive devices such as wheelchairs, walkers, braces, or communication aids may be necessary to help individuals perform daily activities. 6. Lifestyle modifications: Making lifestyle modifications, such as getting enough rest, eating a healthy diet, and avoiding alcohol and drugs, can help support brain health and recovery. 7. Support groups: Joining a support group can provide emotional and social support for individuals with brain injuries and their families. These groups offer opportunities to connect with others who have experienced similar challenges and learn from each other's experiences. 8. Psychological counseling: Psychological counseling, such as cognitive-behavioral therapy or psychotherapy, may be helpful in managing emotional and behavioral changes that can occur after a brain injury. 9. Vocational rehabilitation: Vocational rehabilitation services can help individuals with brain injuries return to work or find new employment opportunities. These services may include job training, education, and counseling. 10. Long-term care: For individuals with severe brain injuries that result in permanent impairment, long-term care may be necessary to manage their daily needs and ensure their overall well-being. This may include home health care, assisted living, or nursing home care.

Analysis: Our response gives us 10 different options for treatment. This does seem like a lot of options and ideally but it does lay out all viable options for the patient so the best decision can be made in this case.

Question Answering using LLM

Query 5 & Analysis

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Answer: First and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions: 1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury. 2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately. 3. Immobilize the leg: Use a splint, sling, or other available materials to immobilize the leg and prevent movement. Be sure not to apply too much pressure on the injury site. 4. Provide pain relief: Offer over-the-counter pain medication, such as acetaminophen or ibuprofen, to help manage pain. 5. Seek medical attention: If the fracture is severe or if you suspect that there may be other injuries, seek medical help as soon as possible. Once you've ensured the person's safety and stability, consider the following steps for their care and recovery: 1. Transportation: Arrange for transportation to a hospital or medical facility as soon as possible. If it's not feasible to move the person, call emergency services for assistance. 2. Diagnosis and treatment: A healthcare professional will diagnose the fracture and determine the best course of treatment based on the severity and location of the injury. This may include immobilization with a cast or brace, surgery, or other interventions. 3. Rehabilitation: After the initial treatment, the person will need to undergo rehabilitation to regain strength, flexibility, and mobility in the affected leg. This may involve physical therapy exercises, assistive devices, and other interventions. 4. Follow-up care: Regular follow-up appointments with a healthcare professional are essential to monitor the healing process and ensure that the fracture is healing correctly. 5. Prevention: Encourage the person to take steps to prevent future injuries while hiking, such as wearing appropriate footwear, using trekking poles, and staying on marked trails.

Analysis: The response goes into the necessary precautions step by step. It then goes into what you should do when it comes to caring for them. This information will be useful to have if you are a civilian and getting someone to the hospital who is suffering from these injuries.

Question Answering using LLM with Prompt Engineering

Prompt and Parameters

The prompt we will use for the same 5 questions will be: “You are an expert medical assistant. Answer the user’s question only from the provided context. If the answer is not in the context, clearly state that you don’t have enough information.”

Our parameters will be: Our max Tokens = 2000, temperature = 0, top_p=0.95, and top_k=50

These are the same parameters as the last 5 questions because we need to compare the responses.

Question Answering using LLM with Prompt Engineering

Query 1 & Analysis

Query 1: What is the protocol for managing Sepsis in a Critical care unit?

Answer: The context does not provide sufficient information to answer the question directly. However, I can share some general guidelines for managing sepsis in a critical care unit based on current medical knowledge and best practices: 1. Early recognition and diagnosis: Sepsis should be suspected in any patient with suspected or confirmed infection and organ dysfunction. Rapid identification and initiation of treatment is crucial to improve outcomes. 2. Fluid resuscitation: Aggressive fluid resuscitation is the first step in managing sepsis. The goal is to maintain adequate tissue perfusion and prevent further organ damage. Crystalloid solutions are typically used initially, with colloids or blood products considered if fluid requirements exceed crystalloid capacity. 3. Vasopressor support: If sepsis-induced hypotension does not respond to fluid resuscitation, vasopressors may be required to maintain adequate mean arterial pressure (MAP) and organ perfusion. Norepinephrine is the first-line vasopressor in most cases. 4. Antibiotics: Broad-spectrum antibiotics should be administered as soon as possible, based on culture results or suspected infection source. 5. Corticosteroids: In some cases of septic shock, corticosteroids may be considered to improve outcomes. However, their use is not universally recommended and should be individualized based on patient factors and clinical judgment. 6. Source control: Addressing the underlying infection source is essential for effective sepsis management. This may involve surgical intervention, drainage, or other interventions depending on the specific infection. 7. Other supportive measures: Additional supportive measures, such as mechanical ventilation, renal replacement therapy, and blood product support, may be required based on the patient's clinical condition. 8. Close monitoring: Septic patients in the critical care unit require close monitoring for hemodynamic instability, organ dysfunction, and infection response to treatment. Adjustments to management strategies should be made as needed based on ongoing assessment.

Analysis:The response does not describe what Sepsis is and gets straight to the point and answers the question and goes into detail for all 8 protocols.

Question Answering using LLM with Prompt Engineering

Query 2 & Analysis

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should follow to treat it?

Answer: Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs include: 1. Abdominal pain: The pain typically starts as a mild discomfort around the navel or in the lower right abdomen. It may then move to the right lower quadrant and become more severe over time. The pain is often described as sharp, crampy, or constant. 2. Loss of appetite: People with appendicitis may lose their appetite due to abdominal discomfort or nausea. 3. Nausea and vomiting: Vomiting is a common symptom of appendicitis, especially in the later stages of the condition. 4. Fever: A fever of 100.4°F (38°C) or higher may be present in some cases of appendicitis. 5. Diarrhea or constipation: Some people with appendicitis experience diarrhea, while others have constipation. 6. Rebound tenderness: When pressure is applied to the abdomen and then released, there may be pain or tenderness in the right lower quadrant, which is a sign of peritonitis (inflammation of the lining of the abdominal cavity). 7. Guarding: People with appendicitis may involuntarily contract their abdominal muscles to protect the area from pressure or pain. Appendicitis cannot be cured via medicine alone, as the inflammation and swelling of the appendix can obstruct the blood flow, leading to tissue death and potential rupture. If left untreated, a ruptured appendix can lead to peritonitis, which is a serious and potentially life-threatening condition. Therefore, surgery is the standard treatment for appendicitis. The surgical procedure used to treat appendicitis is called an appendectomy. During this procedure, the surgeon makes an incision in the abdomen and removes the inflamed appendix. The two most common types of appendectomies are open appendectomy and laparoscopic appendectomy: 1. Open appendectomy: In this procedure, the surgeon makes a single large incision (usually about 2-3 inches long) in the abdomen to access the appendix. The appendix is then removed, and the incision is closed with sutures or staples. 2. Laparoscopic appendectomy: In this minimally invasive procedure, the surgeon makes several small incisions (usually less than 1 inch long) in the abdomen to insert a laparoscope and other surgical instruments. The appendix is then removed using these instruments, and the incisions are closed with sutures or staples. Both open and laparoscopic appendectomies have their advantages and disadvantages, and the choice of procedure depends on various factors, including the patient's age, overall health, and the severity of the appendicitis. In general, laparoscopic appendectomy is preferred when possible due to its faster recovery time and smaller incisions. However, open appendectomy may be necessary in certain cases, such as when the appendix has ruptured or when there are complicating factors that make a laparoscopic procedure more difficult.

Analysis: The response does go in detail about different medications and treatments that can help with the condition. It then talks about the surgical procedures and when each one is the preferred surgery.

Question Answering using LLM with Prompt Engineering

Query 3 & Analysis

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Answer: Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes. The exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certain medications. There are several treatments that have been shown to be effective in addressing sudden patchy hair loss: 1. Corticosteroids: These are anti-inflammatory drugs that can help reduce inflammation and suppress the immune system's attack on the hair follicles. They can be applied topically or taken orally, depending on the severity of the condition. 2. Minoxidil: This is a medication that has been shown to promote hair growth in some people with alopecia areata. It works by increasing blood flow to the affected area and stimulating hair follicles to produce new hairs. 3. DHT blockers: Dihydrotestosterone (DHT) is a hormone that can contribute to hair loss. Some people with alopecia areata have high levels of DHT in their scalp. Finasteride and dutasteride are medications that can block the production of DHT, which may help promote hair growth. 4. Immune system modulators: These drugs work by suppressing the immune system's attack on the hair follicles. Examples include methotrexate, mycophenolate mofetil, and cyclosporine. 5. Hair transplantation: In some cases, hair transplantation may be an option for people with alopecia areata who have large bald patches that don't respond to other treatments. This involves taking healthy hairs from another part of the scalp or body and transplanting them into the affected area. It's important to note that there is no cure for alopecia areata, and not all treatments work for everyone. The effectiveness of treatment can also depend on the severity and location of the bald patches. If you're experiencing sudden patchy hair loss, it's best to speak with a healthcare professional or dermatologist to determine the underlying cause and discuss your treatment options.

Analysis: The response answers the full question and says there is not a known exact cause. It then goes into the different treatments and solution for the patients who have alopecia aerata. It gives us the same answer as the query before prompt engineering

Question Answering using LLM with Prompt Engineering

Query 4 & Analysis

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Answer: A person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are some common treatments that may be recommended: 1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical attention as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening conditions. 2. Medications: Depending on the symptoms, healthcare professionals may prescribe medications to manage various conditions associated with a brain injury, such as pain, swelling, seizures, or infections. 3. Surgery: In some cases, surgery may be necessary to remove blood clots, repair skull fractures, or relieve pressure on the brain. 4. Rehabilitation: Rehabilitation is an essential component of treatment for brain injuries. It may include physical therapy, occupational therapy, speech and language therapy, and cognitive rehabilitation to help individuals regain lost skills and improve overall function. 5. Assistive devices: Depending on the extent of the injury, assistive devices such as wheelchairs, walkers, braces, or communication aids may be necessary to help individuals perform daily activities. 6. Lifestyle modifications: Making lifestyle modifications, such as getting enough rest, eating a healthy diet, and avoiding alcohol and drugs, can help support brain health and recovery. 7. Support groups: Joining a support group can provide emotional and social support for individuals with brain injuries and their families. These groups offer opportunities to connect with others who have experienced similar challenges and learn from each other's experiences. 8. Psychological counseling: Psychological counseling, such as cognitive-behavioral therapy or psychotherapy, may be helpful in managing emotional and behavioral changes that can occur after a brain injury. 9. Vocational rehabilitation: Vocational rehabilitation services can help individuals with brain injuries return to work or find new employment opportunities. These services may include job training, education, and counseling. 10. Long-term care: For individuals with severe brain injuries that result in permanent impairment, long-term care may be necessary to manage their daily needs and ensure their overall well-being. This may include home health care, assisted living, or nursing home care.

Analysis: The response is the same as the original query 4 before prompt engineering

Question Answering using LLM with Prompt Engineering

Query 5 & Analysis

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Answer: First and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions: 1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury. 2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately. 3. Immobilize the leg: Use a splint, sling, or other available materials to immobilize the leg and prevent movement. Be sure not to apply too much pressure on the injury site. 4. Provide pain relief: Offer over-the-counter pain medication, such as acetaminophen or ibuprofen, to help manage pain. 5. Seek medical attention: If the fracture is severe or if you suspect that there may be other injuries, seek medical help as soon as possible. Once you've ensured the person's safety and stability, consider the following steps for their care and recovery: 1. Transportation: Arrange for transportation to a hospital or medical facility as soon as possible. If it's not feasible to move the person, call emergency services for assistance. 2. Diagnosis and treatment: A healthcare professional will diagnose the fracture and determine the best course of treatment based on the severity and location of the injury. This may include immobilization with a cast or brace, surgery, or other interventions. 3. Rehabilitation: After the initial treatment, the person will need to undergo rehabilitation to regain strength, flexibility, and mobility in the affected leg. This may involve physical therapy exercises, assistive devices, and other interventions. 4. Follow-up care: Regular follow-up appointments with a healthcare professional are essential to monitor the healing process and ensure that the fracture is healing correctly. 5. Prevention: Encourage the person to take steps to prevent future injuries while hiking, such as wearing appropriate footwear, using trekking poles, and staying on marked trails.

Analysis: The response goes into the necessary precautions step by step. It then goes into what you should do when it comes to caring for them. This information will be useful to have if you are a civilian and getting someone to the hospital who is suffering from these injuries. This is the same as the original query 5 before prompt engineering.

Data Preparation for RAG

Importing Libraries and Data File

We will import:

Json,os tiktoken and pandas to process data frames and text

We also will import RecursiveCharacterTextSplitter, PyMuPDFLoader, SentenceTransformerEmbeddings and Chroma. These are for loading data, chunking, embedding and Vector Databases

The data file we will use is “medical_diagnosis_manual.pdf” and the length of it is 4114 pages

Data Preparation for RAG

Data Chunking and Embedding

The encoding name is “cl100k_base”. The Chunk_size is 75 and the chunk_overlap is 7

The embedding model name is “all-MiniLM-L6-v2”

The parameters used to define the RAG are k=6, max_tokens=128, temperature=0

Question Answering for RAG

System and User Prompt

System: You are an expert medical assistant. Answer the user’s question only from the provided context. If the answer is not in the context, clearly state that you don’t have enough information.

User Prompt: “””Context: {context}\n\nQuestion: {question}\n\nAnswer:”””

Question Answering using RAG

Query 1 & Analysis

Query 1: What is the protocol for managing Sepsis in a critical care unit?

Answer: In a critical care unit, patients with sepsis should be treated in an ICU and monitored frequently for systemic pressure, central venous pressure (CVP), pulmonary artery occlusive pressure (PAOP) or both, pulse oximetry, arterial blood gases (ABGs), blood glucose, lactate, temperature, and hemodynamic status. Initial therapy should include broad-spectrum antibiotics such as ampicillin plus gentamicin after appropriate cultures are taken. If the patient is in septic shock, they should also receive vasopressors.

Analysis: The response goes into detail but does not drag on a lot. It keeps it short and sweet and gives the best medical advice to the healthcare professional so they can understand the information and not have to read so much. This will help guide the healthcare professional and gives them the best information possible

Question Answering using RAG

Query 2 & Analysis

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should follow to treat it?

Answer: The common symptoms of appendicitis include epigastric or periumbilical pain followed by brief nausea. The condition is typically treated through surgical removal of the appendix, either via open or laparoscopic appendectomy. There is no cure for appendicitis with medication alone.

Analysis: The response is short and sweet. It tells the medical professionals exactly what to do without fluffing it. If the medical professional needs more information they would need to submit another query. I think the response can be more descriptive.

Question Answering using RAG

Query 3 & Analysis

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Answer: The treatment options for sudden patchy hair loss, also known as alopecia areata, include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). The possible causes behind this condition are not well understood but may involve an autoimmune response. It is essential to rule out underlying disorders, such as concomitant virilization in women or scarring.

Analysis: The response gives you all options to help treat alopecia areata. It even tells them to rule out other possible conditions that are similar in nature.

Question Answering using RAG

Query 4 & Analysis

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Answer: The treatment for a person with a brain injury is primarily supportive. This may include infusions of large volumes of isotonic solutions and blood products, cognitive therapy, and rehabilitation services. For patients whose coma exceeds 24 hours, a prolonged period of rehabilitation, particularly in cognitive and emotional areas, may be required. In some cases, medications such as rivastigmine may be used to improve cognition. For individuals with motor neuron disorders, supportive care involves avoiding or modifying activities that cause symptoms, and wrist splints and elbow pads can help reduce pressure and prevent reinjury.

Analysis: The response gives many treatment options but does not go too much into detail. It does mention people who had a coma longer than 24 hours and talks about what may be required for them.

Question Answering using RAG

Query 5 & Analysis

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Answer: For a person who has fractured their leg during a hiking trip, the first step is to provide RICE (Rest, Ice, Compression, Elevation) treatment as soon as possible. This can help reduce swelling and pain. NSAIDs and analgesics may be prescribed for pain relief. If walking is painful, crutches may be required initially. Patients who are elderly or not active with displaced fractures may be treated with prosthetics instead of surgery. Splinting or definitive treatment for certain injuries may also be necessary. After surgery, elevating the legs and avoiding sitting.

Analysis: We receive a surface level response here. It is easy to read and someone who is a medical professional can read this and know exactly what to do step by step.

Question Answering using Fine-Tuning

Query 1 & Analysis

Query 1: What is the protocol for managing Sepsis in a critical care unit?

Answer: In early-onset sepsis, initial therapy should include ampicillin or penicillin G plus an aminoglycoside or another appropriate broad-spectrum antibiotic. Patients with septic shock should be treated in an ICU and their systemic pressure, CVP, PAOP, pulse oximetry, arterial blood gases, blood glucose, lactate levels should be monitored frequently.

Analysis: The parameter used is temperature = 0.5 the reason why I chose this number is so the response can be flexible but also coherent. The response is different than the RAG one because it talks about early-onset sepsis and septic shock.

Question Answering using Fine-Tuning

Query 2 & Analysis

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should follow to treat it?

Answer: The common symptoms of appendicitis include abdominal pain, typically located in the lower right quadrant, loss of appetite, nausea, vomiting, constipation or diarrhea, fever, and a feeling of ill health. Appendicitis cannot be cured via medicine alone; it requires surgical intervention. The most common surgical procedure for treating appendicitis is an appendectomy, which involves the removal of the inflamed appendix.

Analysis: The parameter used is temperature = 0.5 . The response is very descriptive in where the abdominal pain is and lists numerous other symptoms. It then gives clear direction on which surgical procedure is needed. It answers all questions effectively.

Question Answering using Fine-Tuning

Query 3 & Analysis

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

Answer: The possible causes of sudden patchy hair loss, also known as alopecia areata, include no obvious skin or systemic disorders. This condition can be diagnosed through microscopic hair examination or scalp biopsy. Effective treatments for alopecia areata include topical corticosteroids, minoxidil, anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), and psoralen plus ultraviolet A (PUVA). These treatments can help stimulate hair growth and reduce inflammation in the affected area. However, it is essential to note that underlying disorders such as concomitant virilization in women or scarring hair loss should be ruled out through a thorough evaluation before considering these treatments.

Analysis: The parameters used are max_tokens = 300 and temperature = 0.5. This response gives us a warning on the surgical options saying hey are not all extensively studied scientifically so the risk is known and the patient has the best information available to make the decision they want to treat their condition. The response also mentions to rule out any underlying disorders before proceeding with the treatment options for alopecia areata.

Question Answering using Fine-Tuning

Query 4 & Analysis

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

Answer: The treatment for a person with a brain injury is primarily supportive. This includes the administration of large volumes of isotonic solutions and blood products. For patients with severe cognitive dysfunction, extensive cognitive therapy may be required immediately after the injury and continued for months or years. Spinal cord injuries require specific rehabilitation therapy that varies depending on the patient's abnormalities. Support groups may also assist the families of brain-injured patients. For patients whose coma exceeds 24 hours, a prolonged period of rehabilitation, particularly in cognitive and emotional areas, is often required. Rivastigmine, a cholinesterase inhibitor, may improve cognition in some cases. Wrist splints and elbow pads can reduce pressure and prevent reinjury for motor neuron disorders. Surgery is rarely indicated for brain injuries. For patients who cannot be closely observed after discharge, they are generally observed in the emergency department.

Analysis: The parameters we used are max_tokens=300 and temperature 0.5 The response gives many treatment options. It goes into specifics as well when it mentions sever cognitive dysfunction and also spinal cord injuries.

Question Answering using Fine-Tuning

Query 5 & Analysis

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

Answer: For a person who has fractured their leg during a hiking trip, the first priority is to ensure their safety and prevent further injury. This may involve immobilizing the leg with a splint or thigh sleeve and using crutches for support when walking. Rest, ice, compression, and elevation (RICE) should be initiated as soon as possible. Pain relief through NSAIDs and analgesics can also be prescribed if necessary. If the fracture is not displaced and the patient is active, treatment may involve surgical repair through ORIF (open reduction internal fixation). In such cases, early ambulation is encouraged for optimal recovery. For elderly individuals or those who are not active, prosthetic treatment may be considered instead of surgery. During their care and recovery, it's essential to document the patient's symptoms and injuries in detail using their own words, along with any objective documentation such as x-rays or laboratory test results. Any challenges in meeting their care needs should also be documented. Thrombosis prophylaxis may be necessary after surgery. If hemorrhagic shock is suspected, the patient should receive emergency treatment for arterial injuries, which typically involves surgical repair. After surgery, elevating the legs and avoiding prolonged sitting in chairs are recommended to promote proper healing. Cross training and a gradual return to activity can also help with recovery.

Analysis: The parameters we used are max_tokens=300 and temperature 0.5. The response mentions what to do if the patient is elderly or if they are active and what the treatment should be for both groups. It goes into detail for the rehab needed as well.

Output Evaluation

Groundness and Relevance Prompts

Evaluation prompt for groundedness: “You are a helpful assistant that evaluates the ‘groundedness’ of a generated answer. Read the context and the answer below, and determine if the answer is fully supported by the provided context. Respond with ‘yes’ if the answer is sounded in the context, and ‘no’ otherwise.”

Evaluation prompt for relevance: “You are a helpful assistant that evaluates the ‘relevance’ of a generated answer. Read the question, context, and the answer below, and determine if the answer is relevant to the question. Respond with ‘yes’ if the answer is relevant, and ‘no’ otherwise.”

Output Evaluation

Query 1 Evaluation

Evaluation for groundedness: “Yes, the answer is fully supported by the context. The context mentions that critical care medicine includes supportive measures such as respiratory and hemodynamic management, and antibiotic treatment for sepsis and septic shock. It also specifies initial antibiotic therapy for early-onset sepsis with ampicillin or penicillin G plus an aminoglycoside or another antibacterial agent based on culture and sensitivity results. The context also mentions the need to monitor various parameters in patients with sepsis or septic shock, which is consistent with the answer's requirement for frequent monitoring of systemic pressure, CVP, PAOP, pulse oximetry, ABGs, blood glucose, and lactate levels.”

Evaluation for relevance: Yes, the answer is relevant to the question as it outlines the steps for managing sepsis in a critical care unit, which includes initial antibiotic therapy, ICU admission for septic shock patients, and frequent monitoring of various parameters.

Analysis:The evaluation for roundedness goes into a lot of detail and mentions the answer is fully supported by the content. The evaluation for relevance summarizes what the response was and ultimately both responses have a good score.

Output Evaluation

Query 2 Evaluation

Evaluation for groundedness: “Yes, the answer is fully supported by the context. The context mentions that critical care medicine includes supportive measures such as respiratory and hemodynamic management, and antibiotic treatment for sepsis and septic shock. It also specifies initial antibiotic therapy for early-onset sepsis with ampicillin or penicillin G plus an aminoglycoside or another antibacterial agent based on culture and sensitivity results. The context also mentions the need to monitor various parameters in patients with sepsis or septic shock, which is consistent with the answer's requirement for frequent monitoring of systemic pressure, CVP, PAOP, pulse oximetry, ABGs, blood glucose, and lactate levels.”

Evaluation for relevance: Yes, the answer is relevant to the question as it outlines the steps for managing sepsis in a critical care unit, which includes initial antibiotic therapy, ICU admission for septic shock patients, and frequent monitoring of various parameters.

Analysis:The evaluation for roundedness goes into a lot of detail and mentions the answer is fully supported by the content. The evaluation for relevance summarizes what the response was and ultimately both responses have a good score.

Output Evaluation

Query 3 Evaluation

Evaluation for groundedness: Yes, the answer is fully supported by the context. The context mentions alopecia areata as a cause of sudden patchy hair loss and provides some information about its etiology and treatment options. The answer expands on this information by listing specific treatments for alopecia areata and explaining how they work. It also acknowledges that underlying disorders should be ruled out through evaluation, which is consistent with the context's mention of microscopic hair examination or scalp biopsy for definitive diagnosis.

Evaluation for relevance: Yes, the answer is relevant to the question as it provides information on effective treatments and solutions for addressing sudden patchy hair loss, specifically alopecia areata, and possible causes behind it.

Analysis: Our response is relevant and grounded to the original question. The evaluation for groundedness summarizes the entire response and explains why it is a great response.

Output Evaluation

Query 4 Evaluation

Evaluation for groundedness: Yes, the answer is fully supported by the context. The context mentions supportive measures such as infusions of large volumes of isotonic solutions and blood products, cognitive therapy for those with severe cognitive dysfunction, a prolonged period of rehabilitation for individuals whose coma lasts more than 24 hours, and the importance of planning rehabilitation services early. The answer also correctly mentions that spinal cord injury may require specific rehabilitation therapy and that support groups may assist families of brain-injured patients.

Evaluation for relevance: Yes, the answer is relevant to the question as it discusses the recommended treatments for a person with a brain injury, which includes supportive measures such as infusions and blood products, cognitive therapy, and a prolonged period of rehabilitation. The answer also mentions that rehabilitation services should be planned early and that specific rehabilitation therapy may be necessary for individuals with spinal cord injuries.

Analysis: Our response is relevant and grounded to the original question. Our evaluation elaborates why it is a great response.

Output Evaluation

Query 5 Evaluation

Evaluation for groundedness: Yes, the answer is fully supported by the context. The context discusses various treatment options for leg fractures, including RICE principles, surgical repair (ORIF), and prosthetic treatment. The answer summarizes these options and provides additional details on each step of the RICE process. It also mentions the potential need for pain relief medication and crutches, as well as the consideration of age and activity level in determining the best course of treatment. The context also discusses the possibility of hemorrhagic shock due to artery injuries and the need for surgical repair in such cases, which is also mentioned in the answer.

Evaluation for relevance: Yes, the answer is relevant to the question as it provides necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, including RICE principles, potential need for surgery, use of crutches, and consideration for elderly individuals or those with displaced fractures. The answer also mentions the importance of consulting a healthcare professional for individual circumstances.

Analysis: Our response is relevant and grounded to the original question. Our evaluation elaborates why it is a great response.